SmartCropVision – Automated Plant Leaf Disease Detection using CNN

Presented by the Radiant Rebels from Department of Cyber Security.

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Introduction to SmartCropVision

Project Focus

Detecting plant leaf diseases using advanced deep learning techniques, specifically Convolutional Neural Networks (CNN).

Driving Motivation

Addressing significant crop losses faced by farmers due to delayed or inaccurate disease identification.

Real-World Impact

Empowering farmers with early detection through image analysis, facilitating timely treatment, and boosting agricultural yield.

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Addressing the Core Problem in Agriculture

The Challenge: Timely and Accurate Disease Identification

Farmers frequently struggle with identifying plant diseases at early stages, or misdiagnosing them entirely. This often leads to widespread crop destruction before effective interventions can be applied.

- **Time-consuming:** Manual inspection is a laborious process.
- **Requires expertise:** Farmers need deep knowledge to correctly identify diseases.
- **Visual confusion:** Symptoms can be subtle or similar across different diseases.

Why it matters: Early detection mitigates crop loss, enhances food security, and promotes sustainable agricultural practices globally.

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Project Objectives

Our primary goal is to empower farmers and agronomists with an intelligent, user-friendly system for instant disease detection and treatment suggestions.

1 Automated Disease Detection

Develop a Convolutional Neural Network (CNN) model, leveraging MobileNetV2, for precise plant leaf disease identification.

2 High-Accuracy Classification

Implement robust preprocessing and classification of leaf images into various disease categories, ensuring high accuracy.

3 Real-Time Web Application

Construct a responsive Streamlit web application for seamless image upload and analysis by users in real time.

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Core Domains and Algorithms

Project Domain

- Computer Vision
- Deep Learning
- Smart Agriculture
- Artificial Intelligence

Key Algorithms Utilised

- **Convolutional Neural Network (CNN):** Fundamental for automatic feature extraction and classification of plant diseases from leaf images.
- **MobileNetV2:** A highly efficient deep learning architecture, serving as the backbone for transfer learning to optimise performance and reduce training time.
- **ImageDataGenerator (Keras):** Employed for data augmentation techniques (e.g., rotation, zoom, flipping), crucial for improving model generalisation.

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Dataset Insights

Source and Composition

Our comprehensive dataset is compiled from open-source plant disease repositories, including PlantVillage and Kaggle.

- **Classes:** Features over 10 distinct categories, such as Apple Scab, Late Blight, and Healthy leaves, ensuring broad coverage.
- **Images:** Comprises more than 20,000 high-quality leaf images, primarily in JPEG/PNG formats.
- **Standardisation:** All images are uniformly resized to 256x256 pixels for consistent model input.

Preprocessing Steps

- **Normalisation:** Pixel values are normalised to a range of [0, 1] for optimal model performance.
- **Augmentation:** Enhanced with techniques like rotation, zoom, and flipping to improve model robustness.
- **Structuring:** Organised into structured folders for efficient training and testing.

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System Architecture and Workflow

1

Preprocessing

The captured image undergoes essential preprocessing steps, including resizing and normalisation, to prepare it for the CNN model.

2

CNN Model (Trained)

The preprocessed image is fed into our trained CNN model, which performs feature extraction and classification to identify the disease.

3

Image Capture (Leaf)

The process begins with capturing an image of the plant leaf, which will be analysed for disease detection.

4

Prediction Output

The model generates a prediction, such as "Tomato_Blight", indicating the detected disease.

5

Display / Action Taken

The prediction is then displayed via the web user interface, or used to trigger alerts, guiding the user on appropriate actions.

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Tools, Libraries, and Model Training

Model Architecture & Configuration

- **Base Model:** MobileNetV2, pre-trained on ImageNet, for robust feature learning.
- **Input Size:** Standardised at 256 x 256 x 3.
- **Added Layers:** GlobalAveragePooling2D, Dense(128, activation='relu'), and Dense(output_classes, activation='softmax').
- **Optimizer:** Adam, for efficient gradient descent.
- **Loss Function:** Categorical Crossentropy, ideal for multi-class classification.
- **Epochs:** 10, balancing training duration and model performance.
- **Batch Size:** 32, for effective training iteration.

Development Environment

- **Python 3.11:** The primary programming language used.
- **TensorFlow / Keras:** Our chosen deep learning framework for efficient model training.
- **Streamlit:** For developing the interactive web application interface.
- **NumPy:** Essential for numerical computations within the project.

Training Highlights

- **Image Augmentation:** Achieved using ImageDataGenerator, incorporating rotation, zoom, flip, and shear to enhance dataset diversity.
- **Accuracy:** Attained over 97% accuracy on training data, validating model effectiveness.
- **Model Storage:** Final model saved as models/plant_disease_model.h5.

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Results and Evaluation

Model Performance Overview

- **Model Used:** MobileNetV2, pre-trained on ImageNet and fine-tuned for plant disease classification.
- **Input Image Size:** Optimised for 256×256 pixels.
- **Epochs:** Trained over 10 epochs to ensure robust learning.
- **Final Training Accuracy:** Achieved an impressive 87.64%.
- **Final Training Loss:** Recorded at 0.4295, indicating strong model convergence.
- **Evaluation Method:** Predictions rigorously tested on real leaf images via the Streamlit application.
- **Confidence Score:** Demonstrated 100% confidence in multiple test predictions (e.g., Apple Scab, Healthy).

Key Observations

The SmartCropVision model has proven to be highly effective and reliable.

- **Accuracy:** The model consistently and accurately classifies unseen leaf images across diverse disease classes.
- **Confidence:** Predictions consistently return high confidence scores, reinforcing reliability.
- **Dynamic Suggestions:** Treatment recommendations are dynamically displayed based on the detected disease class, providing immediate, actionable insights for farmers.

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Key Takeaways: SmartCropVision

1

High-Accuracy Detection

Successfully developed a deep learning-based system for classifying plant leaf diseases with notable accuracy.

2

Lightweight & Efficient

Utilised MobileNetV2 for fast and effective disease prediction, ensuring a lightweight model ideal for practical applications.

3

Real-Time Interactivity

Created an interactive Streamlit web application, enabling real-time user interaction and immediate prediction results.

4

Strong Confidence

Demonstrated robust model confidence, achieving up to 100% accuracy across various disease categories.

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Thank You

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